



STARTUP VIABILITY INDEX (SVI)

Data Analysis & Visualisation

Development of a composite Startup Viability Index (SVI) to evaluate and compare startup ecosystem conditions across countries

Elga Jerusha Henry

D00262370

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B. Sc. (Hons) Computing in Software Development

GitHub: https://github.com/henryelga/Startup_Viability_Index

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Executive Summary

This study developed the **Startup Viability Index (SVI)**, a composite index designed to evaluate and compare startup ecosystem conditions across countries. The index measures structural factors that influence entrepreneurial activity using four key dimensions:

- Finance and Investment
- Technology and Innovation
- Economic Growth and Stability
- Talent and Productivity

Data were collected primarily from the World Bank and processed through data cleaning, missing value treatment, outlier analysis, normalisation, and multicollinearity testing. Correlation analysis, Variance Inflation Factor (VIF), and Principal Component Analysis (PCA) were used to identify redundancy between indicators and strengthen the framework.

After normalisation, indicators were aggregated into sub-indices using equal weighting and arithmetic mean aggregation. These sub-indices were then combined to calculate the final SVI score for each country.

The results showed clear differences in startup viability across countries. Hong Kong SAR, Singapore, the United States, Korea, Switzerland, and Denmark achieved the highest scores due to strong performance in innovation, financial development, technological infrastructure, and human capital. Lower-ranked countries generally performed weaker across multiple dimensions, particularly in innovation and talent productivity.

Visualisations including histograms, world maps, boxplots, and sub-index comparison charts highlighted regional and income-group patterns. Higher SVI scores were concentrated in North America, Western Europe, and parts of East Asia, while lower-income economies recorded lower overall performance.

The SVI was also compared with the Global Innovation Index (GII) and Global Entrepreneurship Index (GEI). Strong positive Pearson and Spearman correlations supported the reliability and consistency of the framework.

Overall, the Startup Viability Index provides a transparent and multidimensional approach for analysing startup ecosystem conditions and supports comparative analysis, policy evaluation, and future research into entrepreneurial ecosystem development.

1. Introduction

1.1. Background

Startups play a key role in modern economies by driving innovation, creating jobs, and supporting technological development. Countries are increasingly focused on strengthening startup ecosystems to attract entrepreneurs, investors, and skilled workers.

Startup success depends on multiple factors such as funding access, economic stability, technology infrastructure, and workforce quality. A single indicator is therefore insufficient to evaluate startup environments, which motivates the development of a data-driven framework to compare countries.

1.2. Motivation

This project is driven by my interest in startups and entrepreneurship. During my final year project, my team developed a platform called Stockd, which was awarded Best Team Project at Expo 2026 and shortlisted for the Top 50 Student Entrepreneur Awards.

This led me to explore why some countries are more supportive of startups than others and how these conditions can be measured using data, resulting in the Startup Viability Index (SVI).

1.3. Aim and Objectives

The aim is to develop a Startup Viability Index (SVI) to measure how supportive countries are for startup growth and entrepreneurship.

Objectives:

- identify key factors affecting startup success
- analyse international datasets
- apply statistical and visualisation methods
- build a composite index for comparison
- interpret and present results

1.4. Target Audience

The index is intended for entrepreneurs, investors, researchers, and policymakers.

It helps identify countries with strong startup environments and highlights areas needing improvement.

2. Theoretical Framework

2.1. Introduction

The Startup Viability Index (SVI) is based on the idea that startup success depends on multiple interconnected factors rather than a single economic measure. Countries may perform well economically but still lack funding access, technological infrastructure, or skilled labour.

Startup ecosystems therefore vary due to differences in finance, innovation capacity, economic stability, and workforce quality. Traditional indicators such as GDP alone cannot capture these differences, which leads to the need for a multidimensional framework.

This also forms the core problem addressed in this study: existing economic measures are insufficient to fully explain startup ecosystem strength across countries.

To address this, the SVI combines multiple indicators into a composite index following guidance from the OECD Handbook on Constructing Composite Indicators, which recommends grouping related variables into structured sub-indices.

The framework is based on four key dimensions:

- Finance & Investment
- Technology & Innovation
- Economic Growth & Stability
- Talent & Productivity

2.2. Composite Index Approach

A composite index was used because the dataset includes variables measured in different units (percentages, ratios, and monetary values).

This approach allows:

- easier comparison across countries
- identification of strengths and weaknesses
- analysis across multiple dimensions
- simplification of complex datasets

In this project, 16 variables are combined into four sub-indices, which are then aggregated into the final SVI.

2.3. Structure of the Startup Viability Index

The SVI is structured into four sub-indices, each capturing a different dimension of startup ecosystem strength.

Sub-Index	Variables Included
1. Finance & Investment	• Domestic credit to private sector (% of GDP) • Foreign direct investment, net inflows (% of GDP) • Market capitalization of listed domestic companies (% of GDP) • Stocks traded, total value (% of GDP)
2. Technology & Innovation	• Fixed broadband subscriptions (per 100 people) • Individuals using the Internet (% of population) • Secure Internet servers (per 1 million people) • Research and development (R&D) expenditure (% of GDP)
3. Economic Growth & Stability	• GDP growth (annual %) • Inflation, consumer prices (annual %) • Real interest rate (%) • Gross fixed capital formation (% of GDP)
4. Talent & Productivity	• School enrollment, tertiary (% gross) • GDP per person employed (constant 2021 PPP \$) • High-technology exports (% of manufactured exports) • ICT goods imports (% total goods imports)

These sub-indices are combined to form the final index, providing a balanced measure of a country's startup environment.

2.4. Sub-Indices

2.5.1. Finance and Investment

This sub-index measures access to funding and capital, which is essential for startup formation and early growth. Strong financial systems improve access to loans, investment, and capital markets (Trojette, 2016).

Variables used:

- Domestic credit to private sector (% of GDP) - access to bank financing
- Foreign direct investment, net inflows (% of GDP) - international capital inflow
- Market capitalisation of listed companies (% of GDP) - stock market strength
- Stocks traded (% of GDP) - market liquidity and activity.

2.5.2. Technology & Innovation

This sub-index captures digital infrastructure and innovation capacity, both critical for modern startups. Strong ICT systems and R&D investment support business growth and innovation (Leogrande et al., 2022).

Variables used:

- Fixed broadband subscriptions (per 100 people) - internet infrastructure
- Internet usage (% of population) - digital access
- Secure internet servers (per 1 million people) - cybersecurity readiness
- R&D expenditure (% of GDP) - innovation investment.

2.5.3. Economic Growth & Stability

This sub-index measures macroeconomic conditions that affect business confidence and long-term planning. Stable economies support entrepreneurial activity and reduce uncertainty (Arin et al., 2015).

Variables used:

- GDP growth (annual %) - economic expansion
- Inflation rate - price stability
- Real interest rate - cost of borrowing
- Gross fixed capital formation (% of GDP) - investment levels.

2.5.4. Talent & Productivity

This sub-index reflects workforce quality, education, and productivity. Higher skill levels and education are strongly linked to startup success (Ratzinger et al., 2017).

Variables used:

- GDP growth (annual %) - economic expansion
- Inflation rate - price stability
- Real interest rate - cost of borrowing
- Gross fixed capital formation (% of GDP) - investment levels.

2.5. Data Source Selection

All data was sourced from the World Bank Databank due to its reliability, global coverage, and consistent measurement standards.

Using a single source ensures comparability across countries and improves data consistency for later analysis, including normalisation and index construction.

3. Data Selection

3.1. Introduction

This section outlines the data selection process used to construct the Startup Viability Index (SVI). Following the OECD Handbook on Constructing Composite Indicators, variables were selected based on relevance, measurability, country coverage, and analytical value.

The aim was to build a globally comparable dataset capable of evaluating startup ecosystems across countries.

3.2. Dataset Scope

To make the analysis as comprehensive as possible, the dataset was collected across a large time period and country range.

Timeframe

The dataset covers the period 2016-2025, allowing the index to capture long-term trends and major global events such as the COVID-19 pandemic and subsequent recovery.

Country Coverage

The analysis includes 217 countries and territories from the World Bank database, enabling broad global comparison across both developed and developing economies.

3.3. Variable Selection Criteria

Following OECD recommendations, the 16 selected variables were chosen based on four main criteria.

Analytical Soundness

Each variable included in the dataset is supported by existing economic and entrepreneurship research, as discussed in the theoretical framework chapter. The selected indicators are closely linked to startup growth, innovation, investment, economic stability, and workforce quality.

Measurability

The index uses quantitative data rather than subjective survey-based indicators. All variables are numerical and measurable, making them more suitable for statistical analysis and cross-country comparison.

Country Coverage

Variables were selected based on their availability across a large number of countries. This helps reduce excessive data gaps and improves the consistency of the dataset.

Relevance

Each variable represents a specific aspect of startup viability, including access to funding, digital infrastructure, macroeconomic stability, or human capital development.

3.4. Indicator Classification and Direction

Indicators were classified based on their expected impact on startup viability.

Positive indicators increase the overall Startup Viability Index score, while negative indicators reduce it.

Sub-Index	Variable Name	Indicator Type	Direction
Finance & Investment	Domestic credit to private sector (% of GDP)	Input (Capital Availability)	Positive (+)
	Foreign direct investment, net inflows (% of GDP)	Output (Market Attraction)	Positive (+)
	Market capitalization of listed companies (% of GDP)	Input (Financial Market Size)	Positive (+)
	Stocks traded, total value (% of GDP)	Output (Market Liquidity)	Positive (+)
Tech & Innovation	Fixed broadband subscriptions (per 100 people)	Input (Physical Connectivity)	Positive (+)
	Individuals using the Internet (% of population)	Output (Digital Market Reach)	Positive (+)
	Secure Internet servers (per 1 million people)	Input (E-commerce Security)	Positive (+)
	Research and development (R&D) expenditure (% of GDP)	Input (Innovation Pipeline)	Positive (+)
Economic Growth	GDP growth (annual %)	Output (Economic Momentum)	Positive (+)
	Inflation, consumer prices (annual %)	Process (Stability Indicator)	Negative (-)
	Real interest rate (%)	Process (Cost of Capital)	Negative (-)
	Gross fixed capital formation (% of GDP)	Input (Physical Investment)	Positive (+)
Talent & Productivity	School enrollment, tertiary (% gross)	Input (Human Capital Pipeline)	Positive (+)
	GDP per person employed (constant 2021 PPP \$)	Output (Labor Productivity)	Positive (+)
	High-technology exports (% of manufactured exports)	Output (Technical Competitiveness)	Positive (+)
	ICT goods imports (% total goods imports)	Input (Supply Chain Integration)	Positive (+)

Indicators such as inflation and real interest rates were treated as negative variables because higher values may create more difficult conditions for startups through increased uncertainty and borrowing costs.

3.5. Data Source, Collection and Processing

All data was sourced from the World Bank Databank due to its reliability, global coverage, and standardised methodology.

Data was downloaded in CSV format and processed using Python (pandas). Indicator codes were renamed for clarity, and only the 16 selected variables were retained.

Initial preprocessing included converting invalid entries (“.”) into missing values, standardising column names, and ensuring all variables were in numeric format.

3.6. Strengths and Limitations of the Dataset

Strengths

Methodological Consistency

Using the World Bank Databank ensures that variables are collected using standardised methods across countries, allowing fair international comparisons.

Reliable Data Sources

The indicators are compiled from official national statistical offices, central banks, and international organisations, improving the reliability and objectivity of the dataset.

Broad Global Coverage

The inclusion of 217 countries and territories provides a large and diverse dataset for evaluating startup ecosystems globally.

Limitations

Missing Data

Some countries do not report all indicators consistently, leading to gaps in the dataset. Missing data is especially common in developing economies and smaller countries.

Reporting Delays

Certain indicators, particularly Research and Development expenditure, are published with delays, meaning the latest years may contain incomplete records.

Use of Proxy Indicators

Direct startup-specific data is not consistently available across all countries. As a result, broader economic and technological indicators were used as proxy measures for startup viability.

3.7. Data Quality Audit

An initial data quality audit was carried out to measure the level of missing data across each variable.

----- Data Quality Check (Missing Values) -----		
	Missing Records	Missing Percentage (%)
stock_market_liquidity	1530.0	70.51
market_capitalisation	1506.0	69.40
rd_expenditure	1419.0	65.39
real_interest_rate	1179.0	54.33
ict_imports	1106.0	50.97
tertiary_enrollment	1016.0	46.82
credit_to_private_sector	757.0	34.88
high_tech_exports	724.0	33.36
capital_formation	667.0	30.74
labor_productivity	592.0	27.28
inflation_rate	547.0	25.21
internet_usage_rate	477.0	21.98
broadband_subscriptions	461.0	21.24
fdi_inflows	427.0	19.68
gdp_growth	304.0	14.01
secure_servers_density	243.0	11.20

The audit showed that some financial market indicators had high levels of missing data:

- stock market liquidity: 70.51%
- market capitalisation: 69.40%
- R&D expenditure: 65.39%

These gaps are mainly caused by differences in financial market development and national reporting capacity across countries.

In contrast, macroeconomic and digital indicators showed better coverage, such as:

- GDP growth: 14.01% missing
- secure internet servers: 11.20% missing
- internet usage rate: 21.98% missing

The results of the audit confirmed that missing values would need to be addressed before further statistical analysis could be performed.

4. Imputation of Missing Data

4.1. Introduction

Following the OECD Handbook on Constructing Composite Indicators, this section addresses missing values in the dataset prior to index construction.

Handling missing data is essential in composite index development, as incomplete datasets can distort cross-country comparisons and reduce reliability. Since the Startup Viability Index is built using 217 countries over a 10-year period, missing values were unavoidable.

To ensure completeness while maintaining comparability, a structured imputation strategy was applied.

4.2. Data Preparation and Imputation Approach

The dataset was first converted from a yearly panel format into a cross-sectional dataset by selecting the most recent available value for each country-indicator pair. This “latest-value snapshot” approach reduces the impact of reporting delays and ensures the index reflects current conditions.

Remaining missing values were then handled using a two-stage imputation process:

- **Income-group imputation:** Missing values were filled using World Bank income classifications (low, lower-middle, upper-middle, high income). This preserves economic similarity by using peer-country averages rather than a global mean.
- **Fallback global mean imputation:** Any remaining gaps were filled using global averages.

After these steps, the dataset contained no missing values.

4.3. Data Processing Workflow

All processing was implemented in Python using pandas and included:

- converting panel data into a country-level snapshot
- selecting latest available observations
- merging World Bank income group classifications
- applying income-group mean imputation
- applying global mean fallback where required
- performing initial outlier diagnostics using the IQR method

The final dataset was exported for further analysis and index construction.

4.4. Imputation Quality Assessment

To assess reliability, the proportion of imputed values was calculated for each variable.

===== DATA QUALITY REPORT =====		
Variable Name	Imputation Rate (%)	Detected Outliers
broadband_subscriptions	4.1%	0
capital_formation	20.7%	30
credit_to_private_sector	21.2%	5
fdi_inflows	8.8%	40
gdp_growth	2.8%	13
high_tech_exports	17.5%	11
ict_imports	20.7%	10
inflation_rate	12.4%	33
internet_usage_rate	4.6%	1
labor_productivity	18.4%	1
market_capitalisation	59.4%	6
rd_expenditure	41.5%	4
real_interest_rate	38.2%	24
secure_servers_density	0.9%	37
stock_market_liquidity	61.8%	5
tertiary_enrollment	23.0%	1

Low Imputation Variables

Several variables required very little imputation and were largely based on observed data:

- secure_servers_density: 0.9%
- gdp_growth: 2.8%
- broadband_subscriptions: 4.1%
- internet_usage_rate: 4.6%

These indicators had relatively strong global reporting coverage.

Moderate Imputation Variables

Most economic and development indicators fell into a moderate imputation range between approximately 8% and 23%.

Examples include:

- fdi_inflows: 8.8%
- inflation_rate: 12.4%
- high_tech_exports: 17.5%
- tertiary_enrollment: 23.0%

These variables relied partially on peer-group estimation while still maintaining a large proportion of observed values.

High Imputation Variables

Some financial and structural indicators required considerable imputation due to limited reporting availability across countries.

These included:

- `real_interest_rate`: 38.2%
- `rd_expenditure`: 41.5%
- `market_capitalisation`: 59.4%
- `stock_market_liquidity`: 61.8%

Higher missingness in financial indicators reflects structural differences in financial market development and reporting capacity across countries.

4.5. Outlier Analysis

Outliers were identified using the Interquartile Range (IQR) method to detect extreme values that could influence later stages of analysis, particularly normalisation and aggregation.

The analysis showed that several variables contained a higher number of outliers, including FDI inflows (40), secure servers density (37), inflation rate (33), and capital formation (30). These variations mainly reflect genuine structural differences between countries, such as disparities in economic scale, financial development, and digital infrastructure, rather than data errors.

In contrast, some indicators showed very few or no outliers, including broadband subscriptions (0), internet usage rate (1), labour productivity (1), and tertiary enrollment (1), indicating more consistent global distributions.

No outliers were removed from the dataset, as they represent meaningful real-world differences between countries. Instead, their impact is addressed during the normalisation stage of the index construction process.

5. Multivariate Analysis

5.1. Introduction and Methodology

In line with Step 4 of the OECD Handbook on Constructing Composite Indicators, this section examines the internal structure of the dataset before constructing the Startup Viability Index (SVI).

The aim is to assess relationships between indicators, identify redundancy, and reduce dimensionality while ensuring the theoretical framework is supported by the data.

The analysis includes:

- Pearson correlation analysis and heatmaps
- Variance Inflation Factor (VIF) analysis
- Principal Component Analysis (PCA)

Before analysis, all variables were standardised using Z-score normalisation to ensure comparability across different units.

5.2. Pairwise Correlation Results Within Sub-Indices

The visual scatter plots can be found in the notebooks.

Finance & Investment

- Domestic Credit shows moderate positive correlations with Market Capitalisation and Stocks Traded.
- Market Capitalisation and Stocks Traded show a very strong correlation, indicating overlap.
- FDI Inflows show weak relationships with the other indicators.

Technology & Innovation

- Broadband Subscriptions and Internet Usage show a strong positive correlation.
- Both also show moderate relationships with R&D Expenditure.
- Secure Internet Servers show weaker correlations with the remaining variables.

Economic Growth & Stability

- Inflation is negatively related to Real Interest Rates and Capital Formation.
- GDP Growth shows weak relationships with most indicators.
- Overall relationships are weaker than in the other sub-indices.

Talent & Productivity

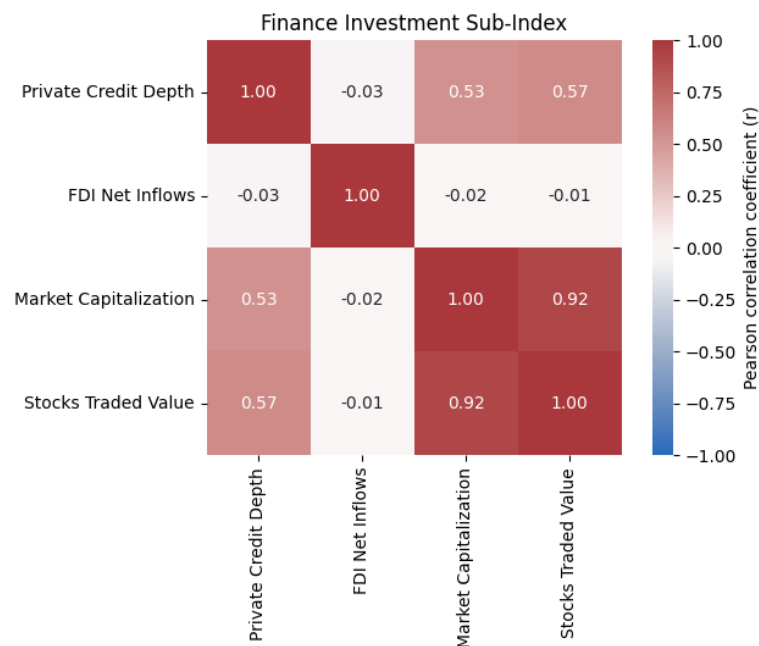
- Tertiary Enrolment and Labour Productivity show a strong positive relationship.
- High-Tech Exports and ICT Imports are also strongly correlated.
- The indicators show generally consistent positive relationships.

5.3. Heatmap Correlation Matrix Analysis

To further evaluate the dataset structure, Pearson correlation heatmaps were generated for each sub-index and for the full 16-variable framework.

The heatmaps make it easier to compare patterns across the full framework rather than only examining pairwise relationships individually.

5.3.1. Finance & Investment Heatmap Findings



The strongest relationship appears between:

- Market Capitalisation and Stocks Traded Value

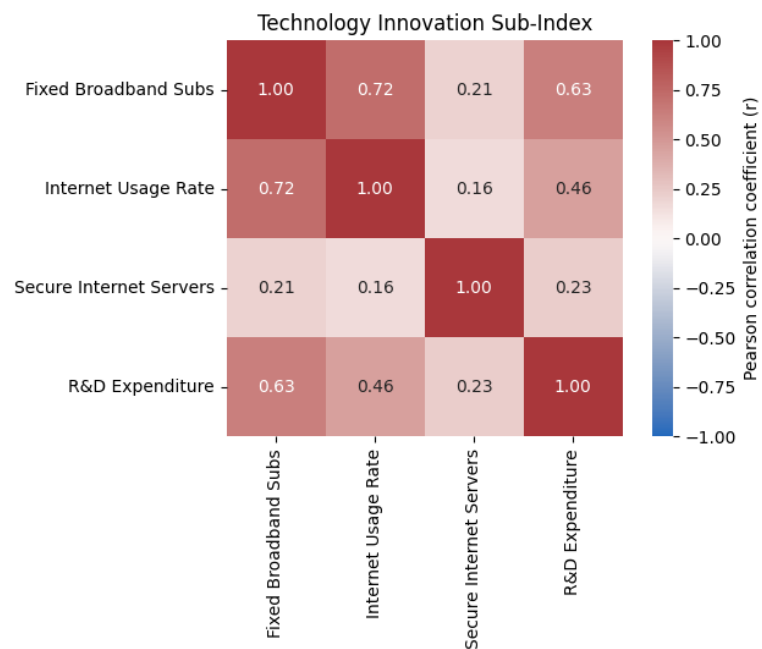
This confirms that both indicators measure very similar stock market conditions.

Moderate positive relationships also appear between:

- Private Credit Depth and Market Capitalisation
- Private Credit Depth and Stocks Traded Value

FDI Inflows remain weakly connected to the other indicators, suggesting that foreign investment operates somewhat independently from domestic financial systems.

5.3.2. Technology & Innovation Heatmap Findings



The Technology & Innovation heatmap shows strong internal consistency.

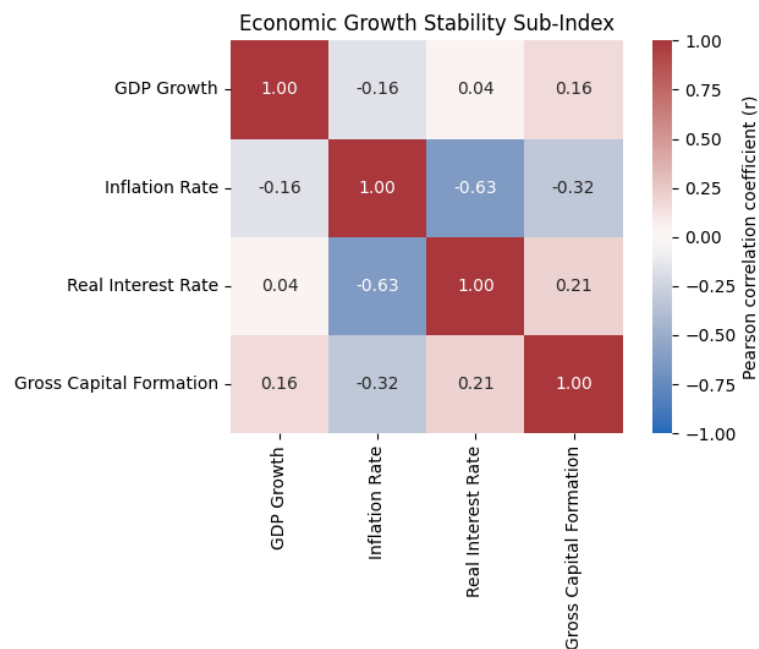
Key relationships include:

- Broadband Subscriptions and Internet Usage
- Broadband Subscriptions and R&D Expenditure

Moderate positive relationships are also visible between Internet Usage and R&D Expenditure.

Secure Internet Servers again show weaker correlations with the rest of the sub-index, indicating that they capture a narrower dimension of digital maturity.

5.3.3. Economic Growth & Stability Heatmap Findings



The Economic Growth & Stability heatmap shows generally weaker relationships.

The strongest relationship appears between:

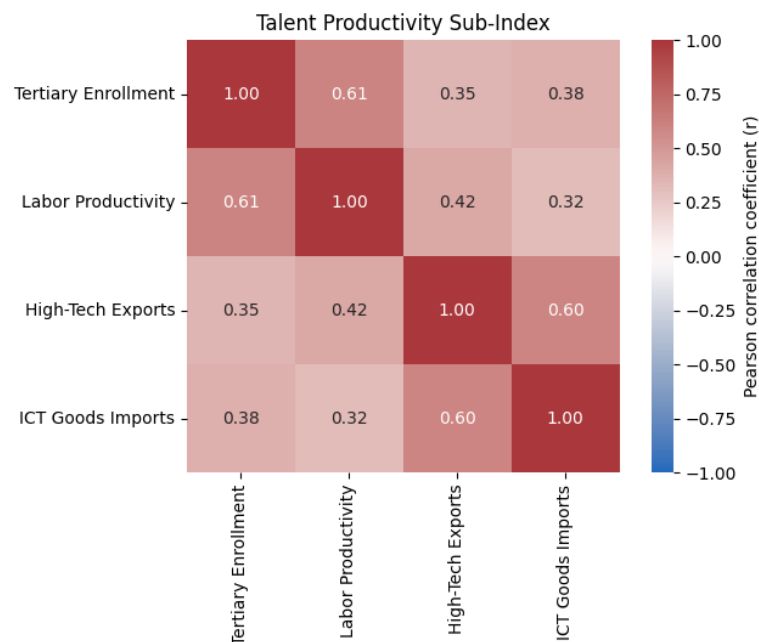
- Inflation Rate and Real Interest Rate

Moderate negative relationships also appear between:

- Inflation Rate and Capital Formation

GDP Growth shows relatively weak relationships with the remaining variables, reinforcing the idea that economic growth behaves independently from several stability indicators.

5.3.4. Talent & Productivity Heatmap Findings



The Talent & Productivity heatmap shows stronger relationships overall.

The clearest relationships include:

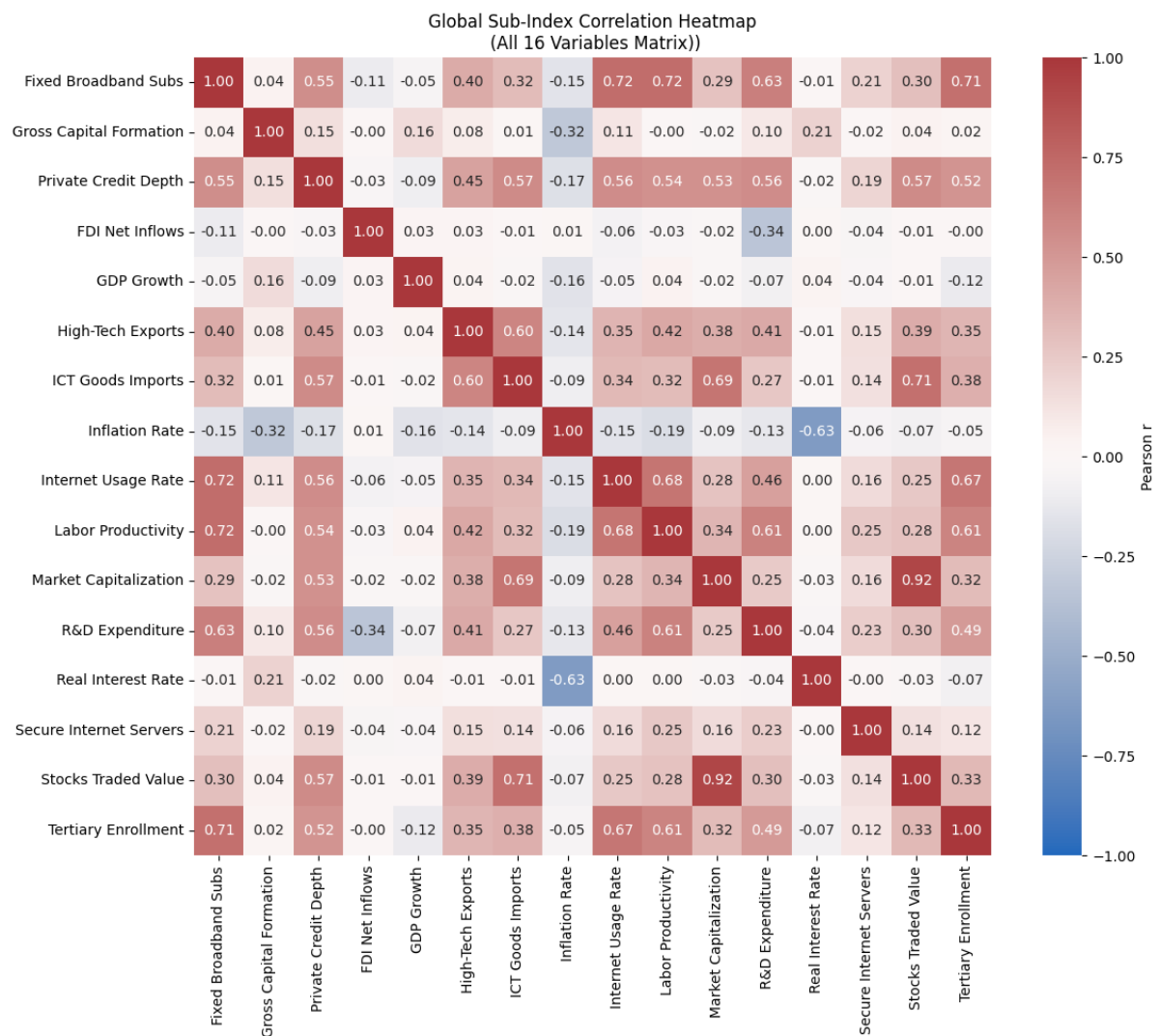
- Tertiary Enrolment and Labour Productivity
- High-Tech Exports and ICT Imports

Moderate positive relationships also appear between:

- Labour Productivity and High-Tech Exports
- Tertiary Enrolment and ICT Imports

These results support the theoretical idea that education, productivity, and technology competitiveness are closely linked within startup ecosystems.

5.3.5. Cross Sub-Index Global Relationships



The global heatmap revealed several important relationships across sub-indices.

Strong positive relationships were observed between:

- Broadband Subscriptions and Labour Productivity
- Internet Usage and Labour Productivity
- Tertiary Enrolment and Broadband Subscriptions

These findings suggest that education, technology access, and productivity are strongly interconnected at the global level.

A moderate negative relationship was observed between:

- FDI Inflows and R&D Expenditure

This suggests that countries investing heavily in domestic innovation do not always rely on large foreign investment inflows.

Overall, the heatmap analysis supports the broader structure of the Startup Viability Index framework.

5.4. Multicollinearity Assessment Using VIF

To formally test redundancy, Variance Inflation Factor (VIF) analysis was applied within each sub-index.

```
Variance Inflation Factor (VIF) Analysis: Sub-Index-Level

>> Sub-Index: Finance & Investment
-----
      Variable Name  VIF Score      Status
credit_to_private_sector    1.4776    Acceptable
      fdi_inflows    1.0013    Acceptable
      market_capitalisation    6.8316 High Multicollinearity
      stock_market_liquidity    7.1919 High Multicollinearity
-----

>> Sub-Index: Technology & Innovation
-----
      Variable Name  VIF Score      Status
broadband_subscriptions    2.7124    Acceptable
      internet_usage_rate    2.0727    Acceptable
      secure_servers_density    1.0624    Acceptable
      rd_expenditure    1.6820    Acceptable
-----

>> Sub-Index: Economic Growth & Stability
-----
      Variable Name  VIF Score      Status
      gdp_growth    1.0485    Acceptable
      inflation_rate    1.8028    Acceptable
      real_interest_rate    1.6675    Acceptable
      capital_formation    1.1322    Acceptable
-----

>> Sub-Index: Talent & Productivity
-----
      Variable Name  VIF Score      Status
tertiary_enrollment    1.6875    Acceptable
      labor_productivity    1.7158    Acceptable
      high_tech_exports    1.6986    Acceptable
      ict_imports    1.6354    Acceptable
-----
```

Most variables showed acceptable levels of multicollinearity ($VIF < 5$), indicating they contribute distinct information.

However, in the Finance & Investment sub-index:

- market_capitalisation (VIF = 6.83)
- stock_market_liquidity (VIF = 7.19)

This confirms high overlap between stock market variables. As a result, stock_market_liquidity was removed prior to PCA to avoid double counting.

All other sub-indices showed low VIF values, confirming strong independence between variables.

5.5. Principal Component Analysis (PCA)

PCA was applied to reduce dimensionality and identify the main underlying structures in the dataset after removing redundant variables.

PCA INITIAL VARIANCE EXTRACTION			
Component	Eigenvalue	Proportion of Variance	Cumulative Variance
PC1	5.0461	0.3349	0.3349
PC2	1.8570	0.1232	0.4581
PC3	1.4528	0.0964	0.5545
PC4	1.1066	0.0734	0.6279
PC5	1.0318	0.0685	0.6964
PC6	0.9541	0.0633	0.7597
PC7	0.8257	0.0548	0.8145
PC8	0.6438	0.0427	0.8572
PC9	0.4887	0.0324	0.8897
PC10	0.3554	0.0236	0.9132
PC11	0.3448	0.0229	0.9361
PC12	0.3213	0.0213	0.9574
PC13	0.2361	0.0157	0.9731
PC14	0.2164	0.0144	0.9875
PC15	0.1887	0.0125	1.0000

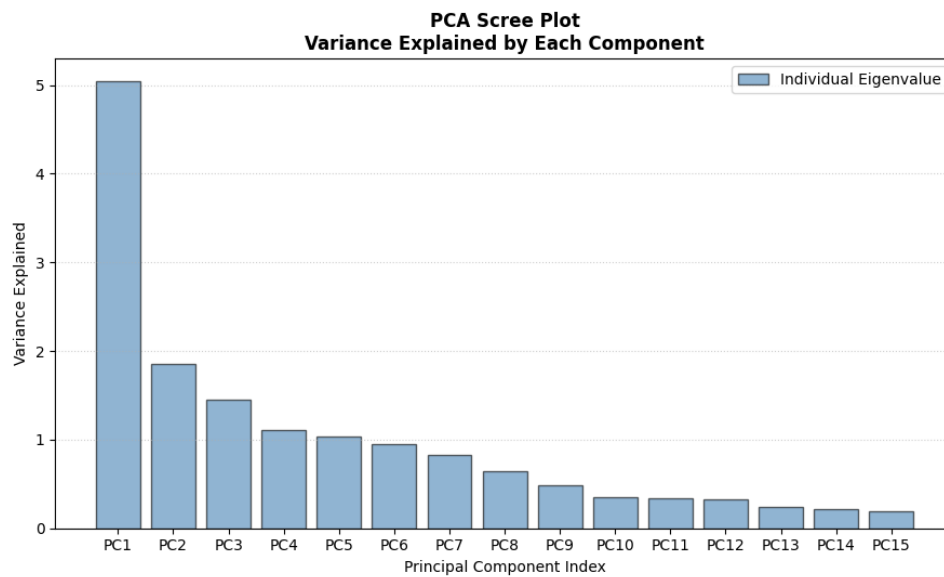
5.5.1. Eigenvalue Analysis (Kaiser Criterion)

Using the Kaiser Criterion, five principal components were retained based on eigenvalues greater than 1, explaining 69.64% of total variance, which is within the accepted range for composite index construction.

- PC1: 33.49%
- PC2: 12.32%
- PC3: 9.64%
- PC4: 7.34%
- PC5: 6.85%

All remaining components explain relatively small amounts of variance and were not retained.

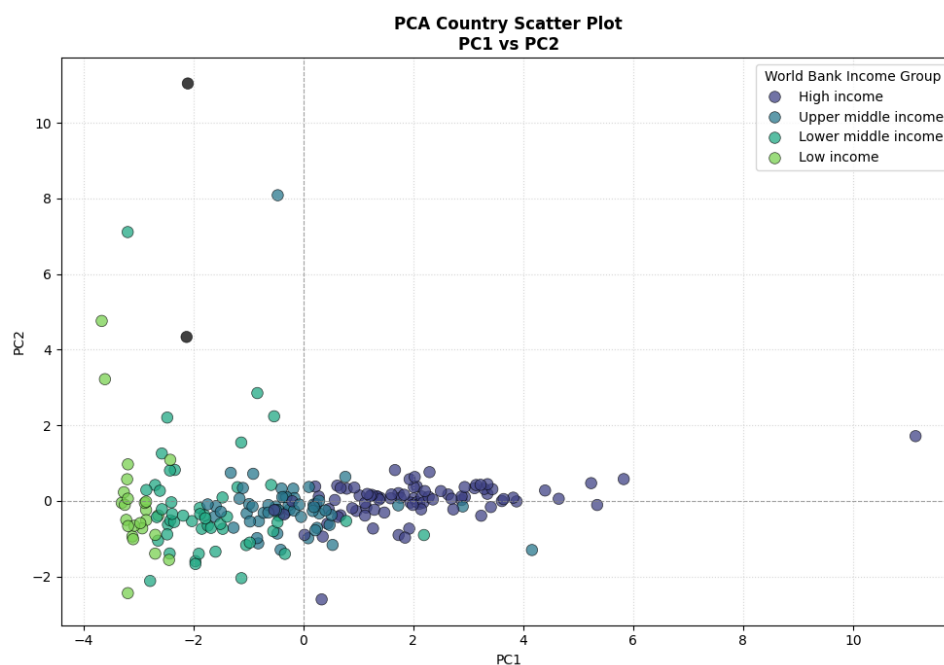
5.5.2. Scree Plot Interpretation



The scree plot shows a sharp decline in eigenvalues from PC1 to PC3, followed by a gradual flattening after PC5. After the fifth component, all eigenvalues fall below 1.0 and contribute limited additional information.

This visual pattern supports the Kaiser Criterion results and confirms that retaining five components is appropriate.

5.5.3. PCA Country Scatter Plot (PC1 vs PC2)



A PCA scatter plot was generated using the first two principal components.

Each point represents a country positioned according to its PC1 and PC2 scores. Countries were coloured using World Bank income group classifications.

The scatter plot shows a clear horizontal separation across PC1:

- High-income countries mostly appear on the right-hand side
- Lower-income countries are concentrated on the left

This suggests that PC1 captures a strong overall development pattern across the dataset.

In contrast, PC2 shows weaker separation between income groups and mainly captures variation within countries rather than broad development differences.

Overall, the scatter plot confirms that the PCA structure reflects meaningful economic differences between countries.

5.5.4. PCA Loading Interpretation

PCA LOADING SCORES (PC1-PC5)					
	PC1	PC2	PC3	PC4	PC5
credit_to_private_sector	0.3565	-0.0016	0.1370	-0.0427	-0.0268
fdi_inflows	-0.0502	-0.0164	0.3143	0.7743	-0.1909
market_capitalisation	0.2590	0.0606	0.4908	-0.1829	-0.0303
broadband_subscriptions	0.3695	0.0298	-0.2698	0.1403	-0.0063
internet_usage_rate	0.3491	0.0001	-0.2156	0.2375	-0.0241
secure_servers_density	0.1334	0.0198	-0.0260	-0.2604	-0.1318
rd_expenditure	0.3242	0.0194	-0.2794	-0.2807	0.1091
gdp_growth	-0.0252	-0.2475	0.0851	0.1496	0.7901
inflation_rate	-0.1019	0.6228	-0.0063	0.0359	0.1689
real_interest_rate	0.0036	-0.5905	0.0019	-0.0651	-0.3961
capital_formation	0.0473	-0.4269	-0.0388	0.0382	0.3106
tertiary_enrollment	0.3419	0.1087	-0.1612	0.2601	-0.0800
labor_productivity	0.3616	0.0001	-0.1916	0.1578	0.0402
high_tech_exports	0.2833	-0.0215	0.3051	-0.0268	0.1336
ict_imports	0.2843	0.0441	0.5281	-0.1410	-0.0012

The PCA loading scores help explain what each principal component represents economically.

- **PC1 (Development & Infrastructure):** driven by broadband access, internet usage, education, productivity, and financial depth
- **PC2 (Economic Stability):** driven by inflation, interest rates, and capital formation
- **PC3 (Technology & Trade Integration):** linked to ICT imports and high-tech exports
- **PC4 (Foreign Investment Dependence):** dominated by FDI inflows
- **PC5 (Short-Term Growth):** primarily driven by GDP growth

5.6. Summary of Multivariate Analysis

The multivariate analysis supports the overall structure of the Startup Viability Index framework.

The results show that:

- most indicators contribute unique information
- relationships within sub-indices are generally consistent
- only one variable showed excessive redundancy
- PCA successfully reduced the framework into five meaningful dimensions

Overall, the dataset demonstrates a stable and interpretable structure suitable for the next stages of normalization, weighting, and composite index construction.

6. Normalisation

6.1. Evaluation and Distribution Assessment

Before constructing the final composite index, the retained indicators were normalised to ensure comparability across different units and scales, including percentages, monetary values, growth rates, and per-capita measures.

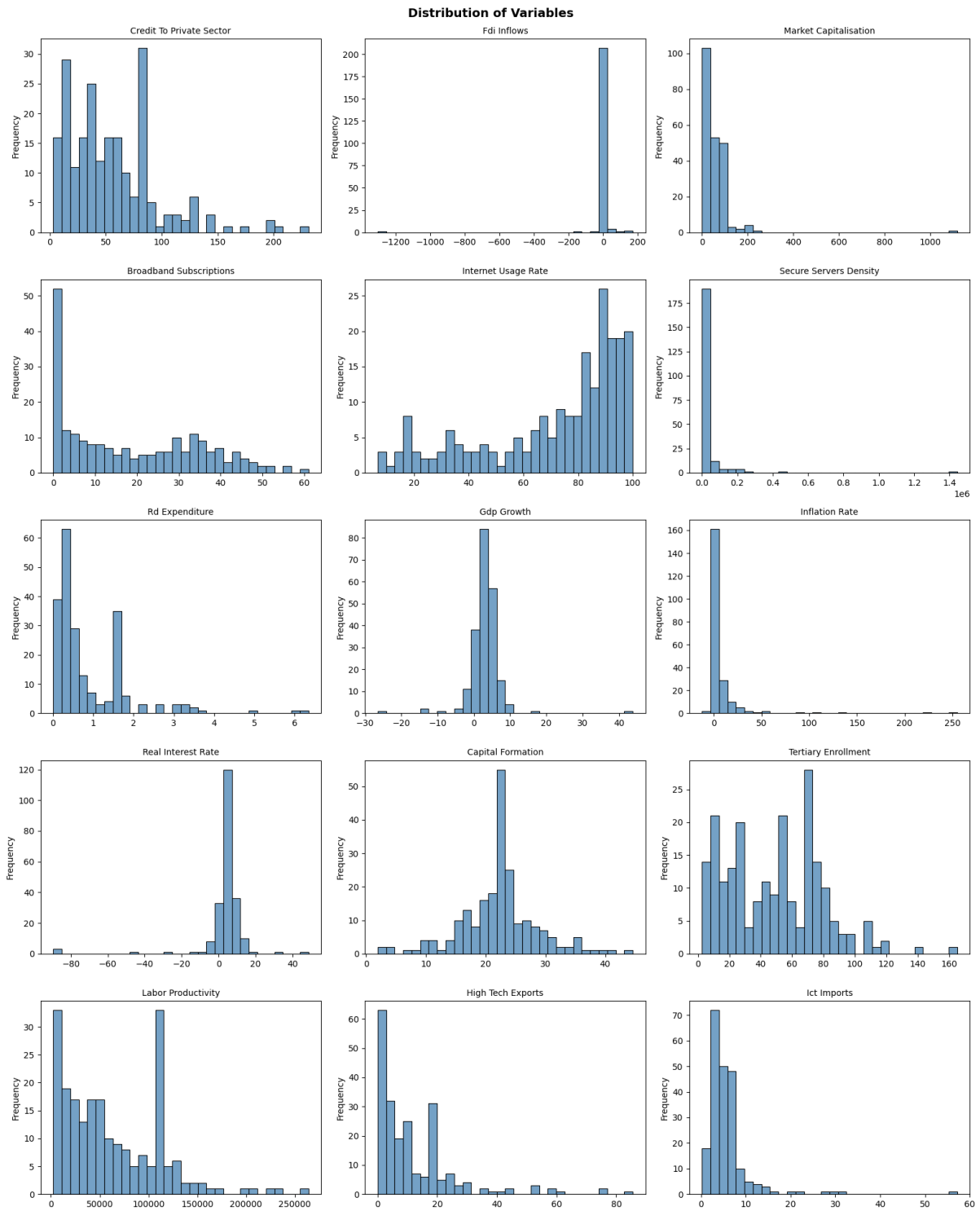
Following OECD guidelines, the indicators were first evaluated for distribution shape, skewness, scale differences, and outlier influence before selecting an appropriate normalisation method.

Several variables showed strong positive skewness, particularly:

- market_capitalisation
- secure_servers_density
- inflation_rate
- ict_imports

Others, including FDI inflows and real interest rates, showed strong negative skewness due to extreme negative observations in some countries.

Histogram analysis confirmed the presence of long tails and outliers across several indicators, indicating that transformation was necessary before aggregation.



6.2. Skewness Reduction

To formally identify problematic distributions, skewness values were calculated for all retained indicators.

Variables Showing High Skewness	
	Skewness
fdi_inflows	-13.5252
market_capitalisation	10.0585
secure_servers_density	10.4109
rd_expenditure	2.2975
inflation_rate	6.7724
real_interest_rate	-5.0407
high_tech_exports	2.3748
ict_imports	5.1318

The following variables exceeded the selected skewness threshold:

- fdi_inflows
- market_capitalisation
- secure_servers_density
- rd_expenditure
- inflation_rate
- real_interest_rate
- high_tech_exports
- ict_imports

A logarithmic transformation was not suitable because several variables contain zero or negative values. Instead, the Yeo-Johnson Power Transformation was applied using Scikit-learn's PowerTransformer.

Unlike logarithmic scaling, the Yeo-Johnson method can handle positive, zero, and negative values while reducing skewness and limiting the influence of extreme outliers. This produced more balanced distributions across the dataset.

6.3. Min-Max Normalisation

After transformation, Min-Max normalisation was applied to rescale all indicators between 0 and 1.

This ensures that variables measured in different units (percentages, monetary values, per-capita measures, growth rates) contribute equally during aggregation and prevents indicators with larger numerical ranges from dominating the final index.

The process preserved relative differences between countries while placing all variables on a common scale.

6.4. Evaluation of Normalised Results

Normalised Results Distribution Summary					
	Minimum	Maximum	Mean	Std Dev	Skewness
credit_to_private_sector	0.0	1.0	0.4611	0.2022	-0.0398
fdi_inflows	0.0	1.0	0.5195	0.0624	1.9015
market_capitalisation	0.0	1.0	0.4658	0.1181	0.0366
broadband_subscriptions	0.0	1.0	0.4767	0.3076	-0.2099
internet_usage_rate	0.0	1.0	0.5866	0.3087	-0.5656
secure_servers_density	0.0	1.0	0.4791	0.2193	0.0048
rd_expenditure	0.0	1.0	0.4647	0.2652	0.2507
gdp_growth	0.0	1.0	0.3821	0.0661	2.6374
inflation_rate	0.0	1.0	0.5940	0.0743	-0.5298
real_interest_rate	0.0	1.0	0.3227	0.0816	2.6305
capital_formation	0.0	1.0	0.4558	0.1503	0.1648
tertiary_enrollment	0.0	1.0	0.4295	0.2105	-0.1373
labor_productivity	0.0	1.0	0.4620	0.2255	-0.1145
high_tech_exports	0.0	1.0	0.4456	0.2370	-0.0108
ict_imports	0.0	1.0	0.5242	0.1420	-0.0545

After applying the Yeo-Johnson transformation and Min-Max normalisation, the indicators become more comparable across countries and less sensitive to extreme outlier effects.

Post-normalisation summary statistics showed that:

- skewness was reduced
- extreme outliers had less influence
- all indicators now shared a common scale
- the dataset became suitable for aggregation into the final composite index

The post-normalisation results show that all indicators were successfully transformed into the expected range between 0 and 1.

Although slight skewness remained in some indicators, the overall distributions were significantly more balanced after transformation and scaling. The final normalised dataset was exported and saved.

This dataset is used in the weighting, aggregation, and final index construction sections.

7. Weighting and Aggregation

7.1. Introduction

After normalisation, the indicators were combined into sub-index scores and the final Startup Viability Index (SVI).

Equal weighting was applied across all indicators within each sub-index, meaning each variable contributes equally to its dimension. This avoids introducing subjective weighting assumptions without strong theoretical justification.

Arithmetic mean aggregation was then used to calculate both sub-index scores and the final composite index. This method is transparent, widely used in composite indicator construction, and allows compensability between indicators, meaning weaker performance in one area can be offset by stronger performance in another.

Potential redundancy between variables had already been assessed through correlation analysis, VIF testing, and PCA. Based on these results, the highly redundant variable `stock_market_liquidity` was removed prior to aggregation.

7.2. Construction of Sub-Indices and Final SVI

Sub-index scores were calculated by taking the arithmetic mean of the normalised indicators within each dimension:

- Finance & Investment
- Technology & Innovation
- Economic Growth & Stability
- Talent & Productivity

Each sub-index therefore represents a country's overall performance within a specific area of the framework.

The final Startup Viability Index (SVI) was then calculated as the arithmetic mean of the four sub-indices:

$$SVI = \frac{Finance + Technology + Economic + Talent}{4}$$

Higher SVI values indicate stronger overall startup ecosystem conditions, while lower values indicate weaker performance relative to other countries.

The completed dataset, including both sub-index scores and final SVI scores, was exported for ranking, visualisation, and further analysis.

7.3. Country Rankings

Top 20 Countries by SVI			Bottom 20 Countries by SVI		
Rank	Country Name	Startup_Viability_Index	Rank	Country Name	Startup_Viability_Index
1	Hong Kong SAR, China	0.737579	198	Comoros	0.300110
2	Singapore	0.689202	199	Somalia, Fed. Rep.	0.298121
3	United States	0.685540	200	Haiti	0.292928
4	Korea, Rep.	0.684858	201	Guinea-Bissau	0.291883
5	Switzerland	0.672583	202	Burkina Faso	0.290679
6	Denmark	0.661012	203	Mozambique	0.289528
7	China	0.659486	204	Zimbabwe	0.285808
8	Norway	0.652871	205	Liberia	0.285746
9	Iceland	0.651246	206	Central African Republic	0.282062
10	Sweden	0.649806	207	Niger	0.279980
11	Australia	0.648738	208	Malawi	0.274455
12	Netherlands	0.644504	209	Eritrea	0.272712
13	Japan	0.643657	210	Burundi	0.272333
14	United Kingdom	0.636734	211	South Sudan	0.271398
15	Cayman Islands	0.636454	212	Yemen, Rep.	0.271160
16	Monaco	0.634251	213	Congo, Dem. Rep.	0.271078
17	Finland	0.626452	214	Sierra Leone	0.267245
18	France	0.623886	215	Chad	0.257280
19	Malta	0.620472	216	Sudan	0.243921
20	Canada	0.619319	217	Afghanistan	0.233031

The highest-ranked countries in the SVI demonstrate strong performance across finance, technology, economic stability, and talent indicators.

Countries such as Hong Kong SAR, Singapore, the United States, and Korea, Rep. achieved particularly high scores due to advanced financial systems, strong digital infrastructure, high levels of innovation, and skilled workforces.

Overall, the results suggest that successful startup ecosystems are typically supported by balanced performance across multiple dimensions rather than strength in a single area alone.

Lower-ranked countries generally recorded weaker performance across several dimensions simultaneously.

Common challenges included limited access to finance, weaker digital infrastructure, lower educational attainment, and reduced investment in innovation and research. In some cases, economic instability also contributed to weaker startup ecosystem conditions.

These results highlight the importance of balanced structural development across all dimensions of the framework.

8. Link to Other Indices

8.1. Introduction

To assess external validity, the Startup Viability Index (SVI) was compared with established global measures of innovation and entrepreneurship.

The objective is not to replicate these indices, but to evaluate whether countries performing strongly on the SVI also tend to rank highly on recognised international benchmarks.

The comparison used:

- the Global Innovation Index (GII), which measures national innovation performance
- the Global Entrepreneurship Index (GEI), which evaluates entrepreneurial ecosystem quality

Both indices are closely related to the innovation, business environment, and productivity dimensions of the SVI.

8.2. Methodology and Data Preparation

Before comparison, several methodological limitations were considered:

- the indices originate from different publication years
- each applies different variables and weighting methods
- country coverage varies across datasets

As a result, the analysis focuses on directional consistency rather than exact numerical equivalence.

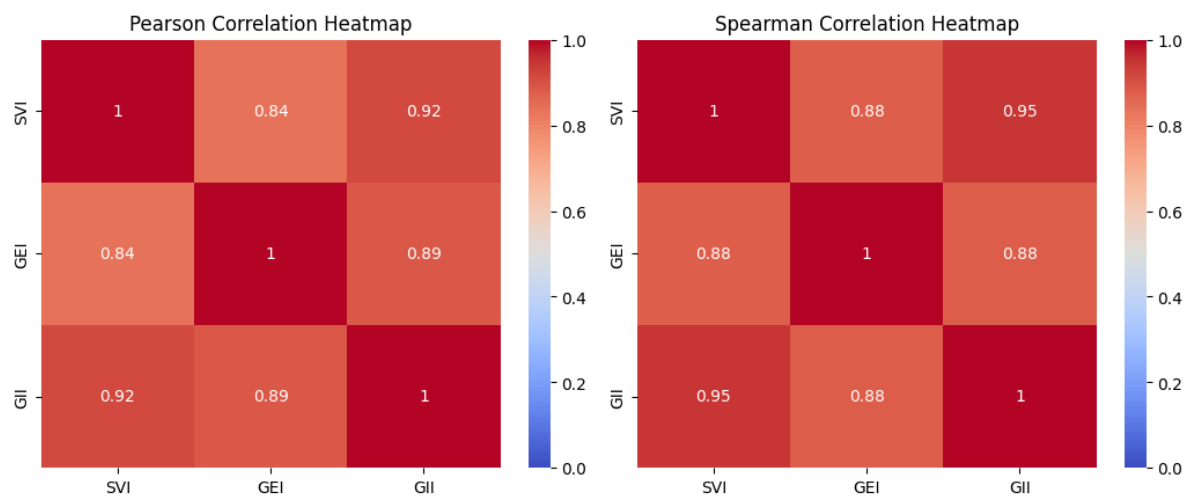
Country names were standardised before merging the SVI, GII, and GEI datasets. After removing missing observations, the final comparison dataset contained 103 countries.

The analysis included:

- Pearson correlation analysis
- Spearman rank correlation analysis
- ranking comparisons and visualisations

Pearson correlation measures linear relationships between scores, while Spearman correlation evaluates consistency in country rankings.

8.3. Correlation Analysis



The results show strong positive relationships between the SVI, GII, and GEI.

Pearson correlations ranged from 0.84 to 0.92, indicating strong linear agreement between the indices. The strongest relationship was observed between the SVI and GII, suggesting that countries with strong innovation systems also tend to exhibit high startup viability.

Spearman correlations were slightly higher, ranging from 0.88 to 0.95, indicating very strong consistency in country rankings even where score magnitudes differed.

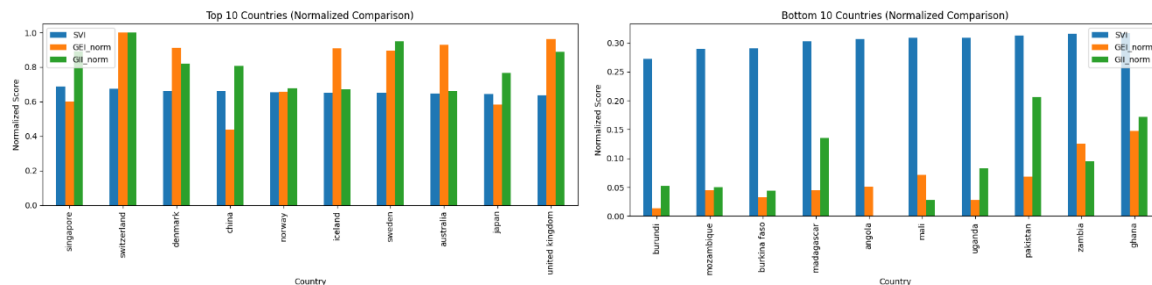
Overall, these findings support the external validity of the Startup Viability Index and demonstrate close alignment with recognised international benchmarks.

8.4. Country Ranking Comparison

To further evaluate the consistency of the Startup Viability Index, country rankings were compared across the SVI, GEI, and GII. This comparison helps determine whether countries that perform strongly on the SVI also tend to rank highly on established innovation and entrepreneurship indices.

Top 10 Countries by SVI				Bottom 10 Countries by SVI			
Country	SVI	GEI	GII	Country	SVI	GEI	GII
singapore	0.689202	0.527036	61.181024	burundi	0.272333	0.118262	13.205343
switzerland	0.672583	0.804472	67.473082	mozambique	0.289528	0.140131	13.109185
denmark	0.661012	0.743097	57.114019	burkina faso	0.290679	0.131650	12.755502
china	0.659486	0.411443	56.322730	madagascar	0.303036	0.140123	17.934865
norway	0.652871	0.566091	49.060133	angola	0.306832	0.144067	10.216209
iceland	0.651246	0.741506	48.543765	mali	0.308710	0.158594	11.794625
sweden	0.649806	0.731051	64.524365	uganda	0.309156	0.128524	14.944522
australia	0.648738	0.754537	48.096089	pakistan	0.312630	0.156412	22.006101
japan	0.643657	0.515181	54.123504	zambia	0.315491	0.195808	15.664772
united kingdom	0.636734	0.777543	61.049318	ghana	0.316882	0.211771	20.034737

Top-ranked countries generally demonstrated strong financial systems, technological infrastructure, innovation capacity, and human capital development. In contrast, lower-ranked countries typically showed weaker performance across these dimensions.



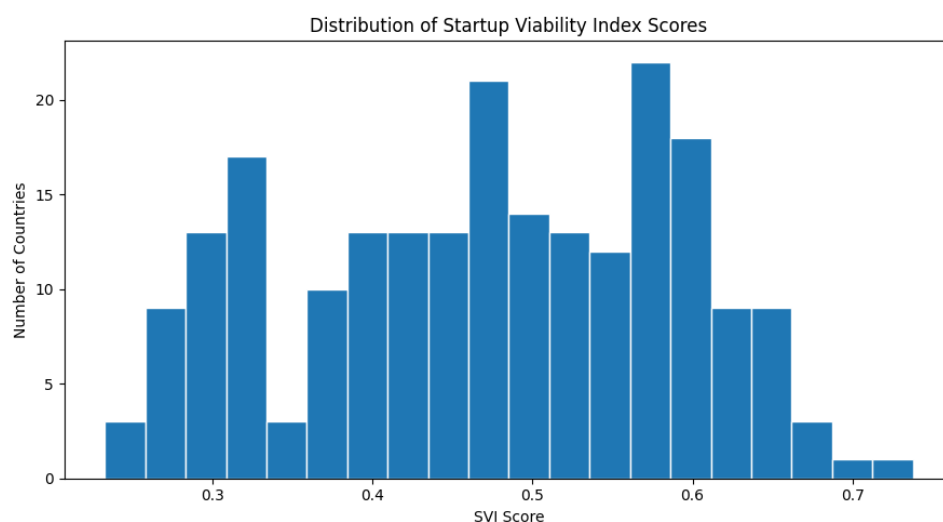
The visual comparisons also revealed some variation among lower-ranked countries, where the SVI occasionally produced stronger results than the GII or GEI. This suggests that the SVI captures broader economic and financial conditions in addition to innovation and entrepreneurship factors.

Overall, the ranking analysis further supports the reliability of the SVI while highlighting its broader multidimensional structure.

9. Visualisation of Results

Visualisations were used to support interpretation of the Startup Viability Index (SVI) and highlight global patterns, regional differences, and variation across the sub-indices.

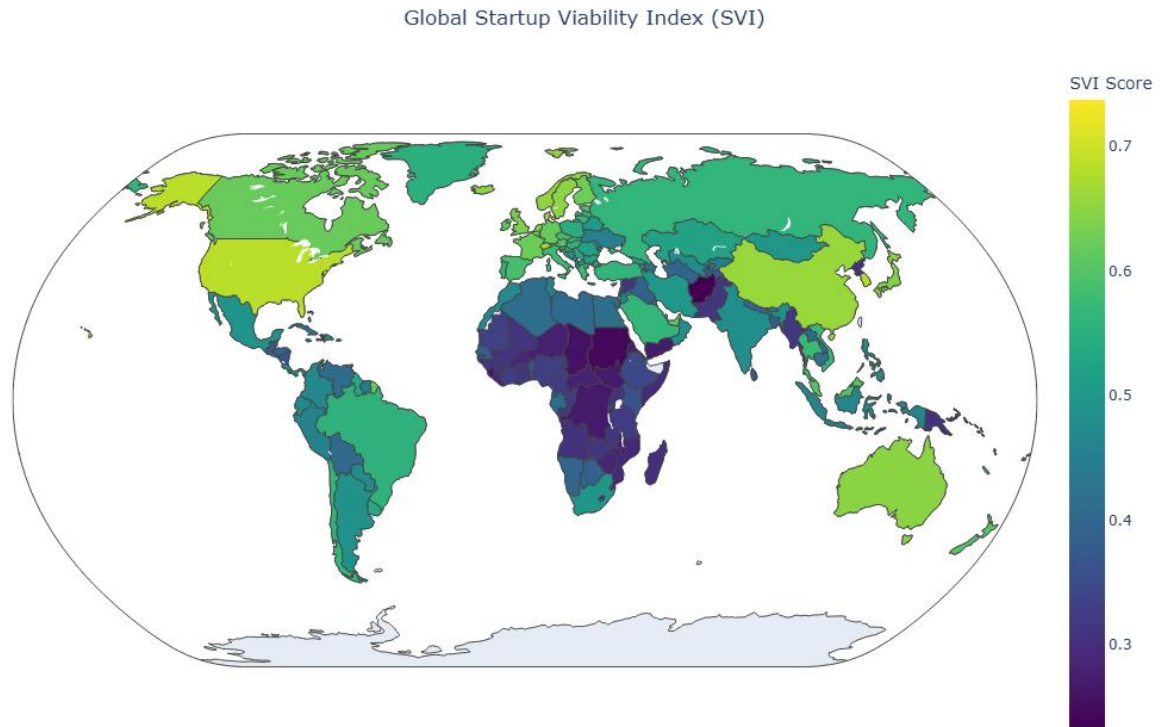
9.1. Distribution of Startup Viability Scores



A histogram was used to examine the distribution of SVI scores across countries.

The results show that most countries are concentrated within the middle range of scores, while relatively few achieve extremely high or low values. This suggests that highly developed startup ecosystems are limited to a smaller group of countries.

9.2. Global Distribution of Startup Viability

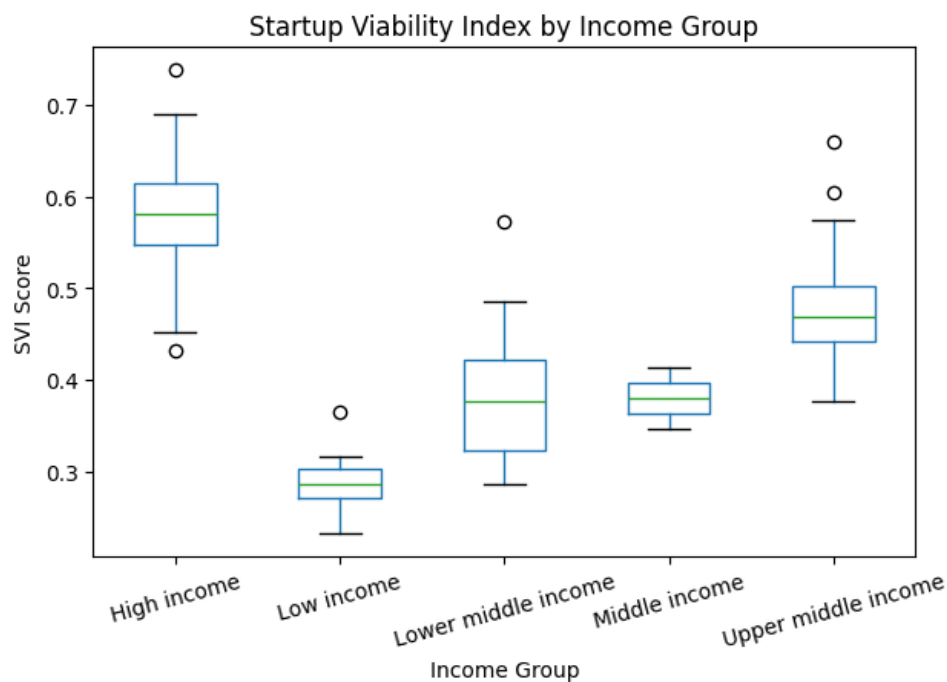


A choropleth world map was used to visualise the geographical distribution of SVI scores.

Higher SVI values are generally concentrated in North America, Western Europe, and parts of East Asia, while lower scores are more common in lower-income and politically unstable regions.

These regional patterns are broadly consistent with differences in economic development, technological infrastructure, institutional quality, and access to investment.

9.3. Startup Viability by Income Group

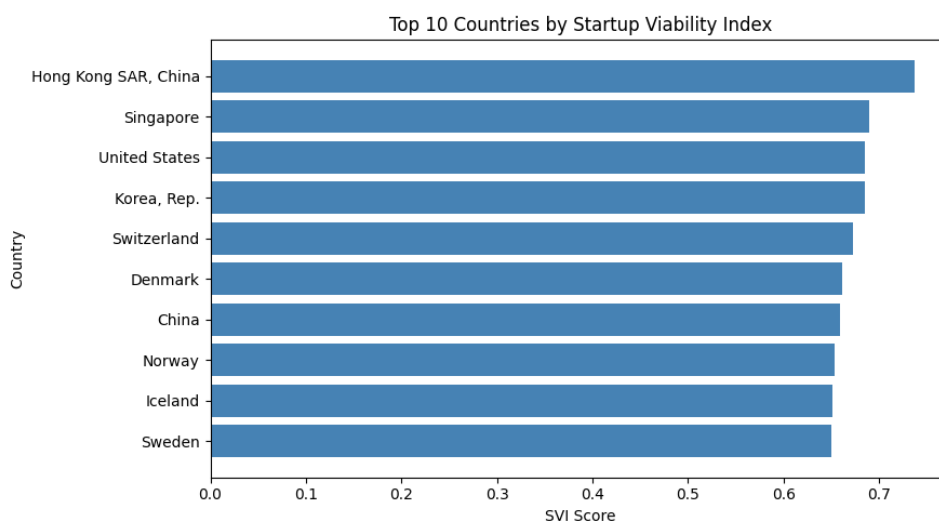


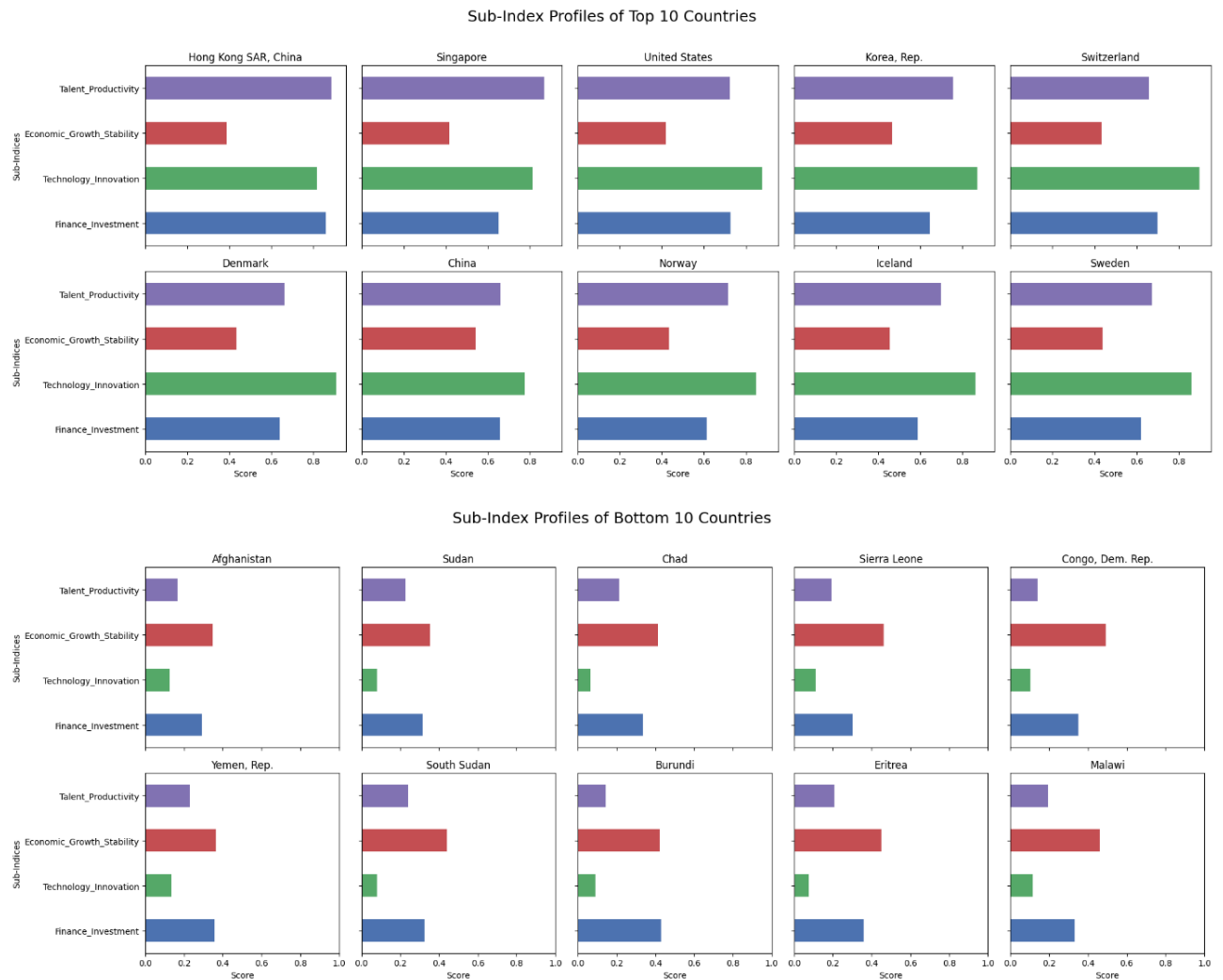
A boxplot was used to compare SVI scores across World Bank income groups.

High-income economies generally recorded higher SVI scores than middle- and low-income countries. Lower-income groups also showed greater variability and lower median performance, indicating wider differences in startup ecosystem development and institutional capacity.

Overall, the results suggest a strong relationship between income level and startup viability.

9.4. Country and Sub-Index Comparisons





Bar charts and sub-index comparison plots were used to examine differences between the highest and lowest-ranked countries.

Top-performing countries consistently showed strong performance across finance, technology, economic stability, and talent dimensions. Many demonstrated particularly strong innovation systems, digital infrastructure, and human capital development.

In contrast, lower-ranked countries generally recorded weaker performance across multiple dimensions, especially in innovation capacity and talent productivity. These countries often face structural constraints such as weaker institutions, limited investment, and lower technological readiness.

The sub-index comparisons also showed that successful startup ecosystems can emerge through different combinations of strengths rather than a single development pathway.

Overall, the visualisations support the validity of the Startup Viability Index and provide a clearer understanding of how startup viability differs across countries and regions.

10. Version Control and Project Structure

Version control was managed using Git throughout the project to ensure structured development, maintainability, and clear tracking of changes.

**Include PCA Loading scores**
henryelga committed yesterday

**Plot PCA Scree Plot and PCA Country Scatter Plot**
henryelga committed yesterday

**Calculate PCA Initial Variance Extraction**
henryelga committed yesterday

**Include VIF Summary and Actions**
henryelga committed yesterday

**Calculate Variance Inflation Factors**
henryelga committed yesterday

**Interpret Heatmap Correlation Matrix**
henryelga committed yesterday

**Generate Correlation Heatmap**
henryelga committed yesterday

☐ 0 Open

☒ 8 Closed

☐  **Visualisation Results**
#8 by henryelga was merged 1 hour ago

☐  **Link to Other Indices**
#7 by henryelga was merged 2 hours ago

☐  **Weighting & Aggregation**
#6 by henryelga was merged yesterday

☐  **Normalisation**
#5 by henryelga was merged yesterday

☐  **Multivariate Analysis**
#4 by henryelga was merged yesterday

☐  **Data Imputation and Outliers Analysis**
#3 by henryelga was merged 2 days ago

Branches

Tags

✓ main default

feature/data-imputation

feature/data-selection

feature/link_other_indices

feature/multivariate_analysis

feature/normalisation

feature/theoretical-framework

feature/visualisation

feature/weighting_aggregation

> artifacts

> data

notebooks

 01_theoretical_framework.ipynb

 02_data_selection.ipynb

 03_data_imputation.ipynb

 04_multivariate_analysis.ipynb

 05_normalisation.ipynb

 06_weighting_aggregation.ipynb

 07_link_other_indices.ipynb

 08_visualisation.ipynb

 main.ipynb

 README.md

Good version control practices were followed, including descriptive commit messages using Gitmoji conventions to improve repository readability and organisation. Separate branches were created for major tasks and sections before being merged into the main branch through pull requests.

The repository also followed a clear folder structure to improve organisation and reproducibility.

This workflow improved project organisation, collaboration, and reproducibility throughout development.

11. Conclusion

11.1. Overview and Main Findings

This study developed the Startup Viability Index (SVI), a composite index designed to evaluate startup ecosystem conditions across countries using four dimensions:

- Finance & Investment
- Technology & Innovation
- Economic Growth & Stability
- Talent & Productivity

The results revealed substantial differences in startup viability between countries. Economies such as Singapore, Switzerland, the United States, and Denmark achieved the highest scores due to strong performance across innovation, finance, technology, and human capital indicators.

In contrast, lower-ranked countries generally showed weaker performance across multiple dimensions, particularly in innovation capacity and talent productivity. The analysis also showed that high-income economies tend to achieve stronger startup ecosystem conditions overall.

11.3. Validation and Limitations

The SVI was compared with the Global Innovation Index (GII) and Global Entrepreneurship Index (GEI) to assess external validity. Strong positive Pearson and Spearman correlations were observed across the indices, supporting the reliability and consistency of the framework.

Several limitations should be acknowledged, including reliance on secondary data sources, the use of equal weighting, and differences in publication years between the external indices used for validation.

Future research could explore alternative weighting approaches, incorporate additional startup-specific indicators, and extend the analysis over time or at regional level.

11.5. Final Conclusion

Overall, the Startup Viability Index provides a transparent and multidimensional framework for analysing startup ecosystem conditions globally.

The findings demonstrate that startup viability is strongly associated with innovation capacity, financial development, economic stability, and human capital. The index therefore offers a useful tool for comparative analysis, policy evaluation, and future research into entrepreneurial ecosystem development.

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